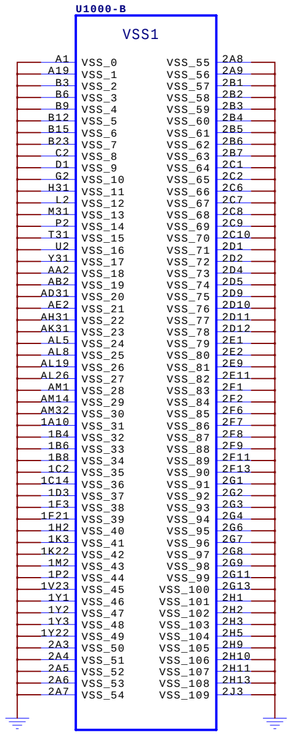
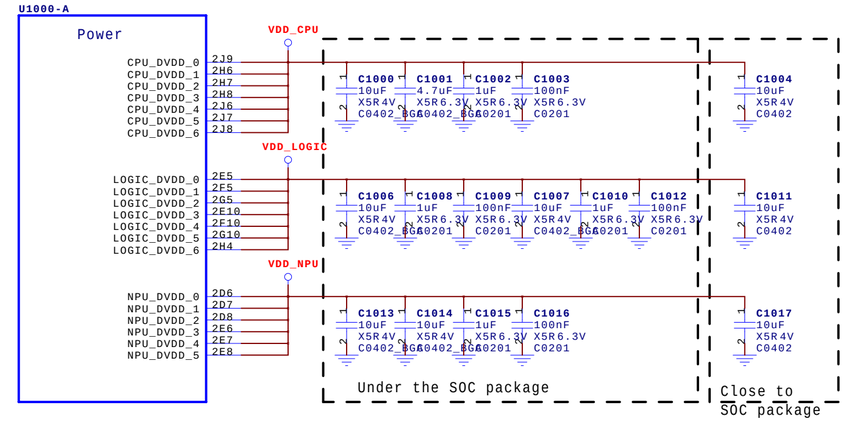
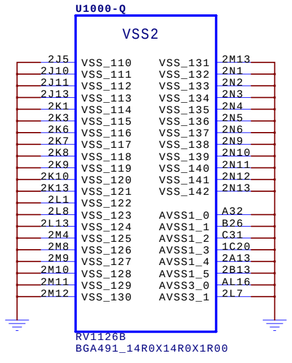
GND



Power





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OSC/PLL/RTC/OTP/PMU

U1000-H

OSC/PLL

PMUIO0&PMUIO1 Domain Logic Power

R1101 22R R0402 — OSC_XOUT — AL14 — OSC_XOUT

Y1100 — 24MHz CRY4_2R50X2R00X0R32

R1103 1M R0402

OSC_XIN — AL15 — OSC_XIN

C1102 12pF C0G50V C0402

C1103 12pF C0G50V C0402

VDD_LOGIC

2K4 — PLL_AVDD_0V9

C1105 1uF X5R6.3V C0201

VCC_1V8

2J4 — PLL_AVDD_1V8

C1106 100nF X5R6.3V C0201

RTC

C1107 18pF C0402 C0G50V — RTC_XIN — AM8 — OSC_RTC_XIN

Y1101 — 32.768KHz CRY2_3R20X1R50X0R90

C1109 18pF C0402 C0G50V — RTC_XOUT — AM7 — OSC_RTC_XOUT

VBAT_RTC

R1124 100R R0402

VCC_RTC

2M3 — RTC_AVDD

Vmax=3.6V

C1110 1uF X5R6.3V C0201

OTP

VCC_1V8

2C13 — OTP_VCC_1V8

C1111 100nF X5R6.3V C0201

TSADC

SARADC1&2

2H12 — TSADC_SARADC1&2_AVDD_1V8

C1603 100nF X5R6.3V C0201

PMUIO_VDD0V9_____OSC_VDD_0V9

LDO

PMUIO_3V3

PMUI00 Domain

Operating Voltage=3.3V

TEST_CLK0_OUT___REF_CLK0_OUT____GPIO0_A0

TSADC_SHUT_____TSADC_SHUTORG___PWM1_CH1

RTC_32K_OUT____CLK_32K_______GPIO0_A2

PWR_CTRL0_____GPIO0_A3_U

PWR_CTRL1_____GPIO0_A4_D

PWM1_CH0_M0____SDMMC0_DET_____GPIO0_A5

SPI0_CS1N_M0____FSPI1_D2_M0____GPIO0_A6

SPI0_CS0N_M0____FSPI1_CSN0_M0___GPIO0_A7

SPI0_MOSI_M0____FSPI1_D0_M0____GPIO0_B0

SPI0_MISO_M0____FSPI1_D1_M0____GPIO0_B1

SPI0_CLK_M0_____FSPI1_CLK_M0____GPIO0_B2

JTAG_TCK_M0_____I2C1_SCL_M0____PWM1_CH2

JTAG_TMS_M0_____I2C1_SDA_M0____PWM1_CH3

PMUIO0_VCC3V3

PMUI01 Domain

Operating Voltage=1.8/3.3V

I2C3_SCL_M0____PWM0_CH7_M0

I2C3_SDA_M0____PWM0_CH6_M0____PWR_CTRL

I2C0_SCL_M0_____SPI2AHB

I2C0_SDA_M0_____SPI2AHB

UART1_TX_M0_____I2C5_SCL_M0____PWM0_CH0

UART1_RX_M0_____I2C5_SDA_M0____PWM0_CH1

UART1_RTSN_M0_____PWM0_CH2

UART1_CTSN_M0_____PWM0_CH3

I2C2_SCL_M0____PWM0_CH4_M0

I2C2_SDA_M0____PWM0_CH5_M0

PMUIO1_VCC

2K5 VDD0V9_PMUIO — C1100 C0201 100nF X5R6.3V

VCC3V3_SYS

2L5 — C1101 100nF X5R6.3V C0201

R1104 1K R0402

NPOR_DET — AL13 — RESET

C1104 1nF X5R25V C0201

AM13 — SPK_CTRL

AL11 — TSADC_SHUT — R1102 0R R0402

AL12 — CLK_32K_OUT

AM11

1AB6 — WIFI_PWREN_L

AL10 — SDMMC0_DET

1AA6 — SDMMC0_VOL_CTRL

1AC4 — SENSOR_INT

1AB4 — SDMMC0_PWREN

1AA4 — WIFI_WAKE_HOST_3V3

1AC2 — WIFI_REG_ON_3V3

AM10 — UART0_TX_DBG

AL9 — UART0_RX_DBG

VCC3V3_SYS

2L4 — C1108 100nF X5R6.3V C0201

1AC12 — PMIC_INT

1AC10 — PWR_CTRL

1AA12 — I2C0_SCL_PMIC

1AB12 — I2C0_SDA_PMIC

1AC8 — PWM_CPU

1AA10 — USBCC_INT_L

1AB10 — LED_PWM_WHITE

1AB8 — LED_PWM_IR

1AA8 — I2C2_SCL_SENSOR

1AC6 — I2C2_SDA_SENSOR

VCC3V3_SYS

2M5 — C1112 100nF X5R6.3V C0201

VCC3V3_SYS

R1135 4.7K R0402 — I2C2_SCL_SENSOR

R1136 4.7K R0402 — I2C2_SDA_SENSOR

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DDR Controller

U1000-C

DDR3	DDR4	LPDDR3	LPDDR4/4x		LPDDR4/4x	LPDDR3	DDR4	DDR3		
									A13	LPDDR3_DDR3_A0
LPDDR3_D7	R1	DDR3_DQ0	DDR4_DQL0_A	LPDDR3_D	LPDDR4_ODT1_B	LPDDR3_A0	DDR4_ODT		A10	LPDDR3_DDR3_A1
LPDDR3_D6	1M3	DDR3_DQ1	DDR4_DQL1_A	LPDDR3_D	LPDDR4_CKE0_A	LPDDR3_A1	DDR4_A8_		1C12	LPDDR3_DDR3_A2
LPDDR3_D4	R2	DDR3_DQ2	DDR4_DQL2_A	LPDDR3_D	LPDDR4_ODT1_A	LPDDR3_A2	DDR4_BG1		B13	LPDDR3_DDR3_A3
LPDDR3_D0	1H1	DDR3_DQ3	DDR4_DQL3_A	LPDDR3_D	LPDDR4_CKE1_A	LPDDR3_A3	DDR4_A0_		1A8	LPDDR3_DDR3_A4
LPDDR3_D3	T2	DDR3_DQ4	DDR4_DQL4_A	LPDDR3_D	LPDDR4_ODT0_A	LPDDR3_A4	DDR4_A7_		A11	LPDDR3_DDR3_A5
LPDDR3_D7	T1	DDR3_DQ5	DDR4_DQL5_A	LPDDR3_D	LPDDR4_A5_A	LPDDR3_A5	DDR4_A9_		1B10	LPDDR3_DDR3_A6
LPDDR3_D5	1H3	DDR3_DQ6	DDR4_DQL6_A	LPDDR3_D	LPDDR4_CLKP_A	LPDDR3_A6	DDR4_A4_		1A12	LPDDR3_DDR3_A7
LPDDR3_D1	1P1	DDR3_DQ7	DDR4_DQL7_A	LPDDR3_D	LPDDR4_A2_A	LPDDR3_A7	DDR4_A11		1C10	LPDDR3_DDR3_A8
					LPDDR4_CLKN_A	LPDDR3_A8	DDR4_A10		1B12	LPDDR3_DDR3_A9
LPDDR3_DM0	1M1	DDR3_DM0	DDR4_DML_A	LPDDR3_D	LPDDR4_A4_A	LPDDR3_A9	DDR4_A2_		1A4	
					LPDDR4_A0_B	---	DDR4_A15		B7	
LPDDR3_DQS0P	1K1	DDR3_DQS0P	DDR4_DQSL_P_A	LPDDR3_D	LPDDR4_CSN0_B	---	DDR4_A16		A8	
LPDDR3_DQS0N	1K2	DDR3_DQS0N	DDR4_DQSL_N_A	LPDDR3_D	LPDDR4_ODT0_B	---	DDR4_A5_		B14	
					LPDDR4_A3_A	---	DDR4_A13		A14	
LPDDR3_D10	M2	DDR3_DQ8	DDR4_DQU0_A	LPDDR3_D	LPDDR4_CSN1_B	---	DDR4_A14		1A6	
LPDDR3_D11	M1	DDR3_DQ9	DDR4_DQU1_A	LPDDR3_D	LPDDR4_A5_B	---	DDR4_A3_			
LPDDR3_D9	N1	DDR3_DQ10	DDR4_DQU2_A	LPDDR3_D						
LPDDR3_D8	N2	DDR3_DQ11	DDR4_DQU3_A	LPDDR3_D	---	---	DDR4_CSN		B10	
LPDDR3_D12	H2	DDR3_DQ12	DDR4_DQU4_A	LPDDR3_D	LPDDR4_CKE0_B	---	DDR4_A1_		1C8	
LPDDR3_D13	J2	DDR3_DQ13	DDR4_DQU5_A	LPDDR3_D	LPDDR4_A0_A	---	DDR4_A6_		B8	
LPDDR3_D14	J1	DDR3_DQ14	DDR4_DQU6_A	LPDDR3_D						
LPDDR3_D15	K1	DDR3_DQ15	DDR4_DQU7_A	LPDDR3_D	LPDDR4_A1_A	---	DDR4_A12		1C4	
					LPDDR4_A1_B	---	DDR4_CKE		B4	
LPDDR3_DM1	K2	DDR3_DM1	DDR4_DMU_A	LPDDR3_D	LPDDR4_CKE1_B	---	DDR4_BA1		1C6	
LPDDR3_DQS1P	1F1	DDR3_DQS1P	DDR4_DQSU_P_A	LPDDR3_D	LPDDR4_CLKP_B	LPDDR3_CLKP	DDR4_CLK		1B2	LPDDR3_DDR3_CLKP
LPDDR3_DQS1N	1F2	DDR3_DQS1N	DDR4_DQSU_N_A	LPDDR3_D	LPDDR4_CLKN_B	LPDDR3_CLKN	DDR4_CLK		1A2	LPDDR3_DDR3_CLKN
LPDDR3_D16	Y2	DDR3_DQ16	DDR4_DQL0_B	LPDDR3_D	LPDDR4_A3_B	LPDDR3_CKE	DDR4_ACT		A5	LPDDR3_DDR3_CKE
LPDDR3_D17	W1	DDR3_DQ17	DDR4_DQL1_B	LPDDR3_D	LPDDR4_A2_B	LPDDR3_ODT0	DDR4_CSN		A4	LPDDR3_DDR3_ODT0
LPDDR3_D23	V1	DDR3_DQ18	DDR4_DQL2_B	LPDDR3_D	LPDDR4_CSN1_A	LPDDR3_ODT1	DDR4_BG0		A7	
LPDDR3_D22	V2	DDR3_DQ19	DDR4_DQL3_B	LPDDR3_D	LPDDR4_A4_B	LPDDR3_CSN0	DDR4_ODT		B5	LPDDR3_DDR3_CS0N
LPDDR3_D21	AE1	DDR3_DQ20	DDR4_DQL4_B	LPDDR3_D	LPDDR4_CSN0_A	LPDDR3_CSN1	DDR4_BA0		B11	LPDDR3_DDR3_CS1N
LPDDR3_D18	AD1	DDR3_DQ21	DDR4_DQL5_B	LPDDR3_D						
LPDDR3_D17	AD2	DDR3_DQ22	DDR4_DQL6_B	LPDDR3_D	LPDDR4_RESETN	---	DDR4_RES		2A1	
LPDDR3_D20	AC2	DDR3_DQ23	DDR4_DQL7_B	LPDDR3_D						
							DDR_RZQ		2A2	DDR_RZQ
LPDDR3_DM2	W2	DDR3_DM2	DDR4_DML_B	LPDDR3_D						
LPDDR3_DQS2P	AB1	DDR3_DQS2P	DDR4_DQSL_P_B	LPDDR3_D						
LPDDR3_DQS2N	AA1	DDR3_DQS2N	DDR4_DQSL_N_B	LPDDR3_D						
							DDR_VREF_OUT		2J1	
LPDDR3_D30	C1	DDR3_DQ24	DDR4_DQU0_B	LPDDR3_D						
LPDDR3_D31	B1	DDR3_DQ25	DDR4_DQU1_B	LPDDR3_D			DDR_VDDQ_0		2C3	
LPDDR3_D28	A2	DDR3_DQ26	DDR4_DQU2_B	LPDDR3_D			DDR_VDDQ_1		2C4	
LPDDR3_D29	B2	DDR3_DQ27	DDR4_DQU3_B	LPDDR3_D			DDR_VDDQ_2		2C5	
LPDDR3_D24	G1	DDR3_DQ28	DDR4_DQU4_B	LPDDR3_D			DDR_VDDQ_3		2D3	
LPDDR3_D25	F1	DDR3_DQ29	DDR4_DQU5_B	LPDDR3_D			DDR_VDDQ_4		2E3	
LPDDR3_D26	F2	DDR3_DQ30	DDR4_DQU6_B	LPDDR3_D			DDR_VDDQ_5		2F3	
LPDDR3_D27	E2	DDR3_DQ31	DDR4_DQU7_B	LPDDR3_D						
							DDR_VDDQL_0		2E4	
LPDDR3_DM3	D2	DDR3_DM3	DDR4_DMU_B	LPDDR3_D			DDR_VDDQL_1		2F4	
LPDDR3_DQS3P	1D1	DDR3_DQS3P	DDR4_DQSU_P_B	LPDDR3_D						
LPDDR3_DQS3N	1D2	DDR3_DQS3N	DDR4_DQSU_N_B	LPDDR3_D						

DDR Power

	VDDQ	VDDQL
DDR3L	1.35V	1.35V
DDR3	1.50V	1.50V
DDR4	1.20V	1.20V
LPDDR3	1.20V	1.20V
LPDDR4	1.10V	1.10V
LPDDR4x	1.10V	0.60V

R1201 120R 0201 1%

VREFOUT_DDR

C1200 100nF C0201 X5R 6.3V

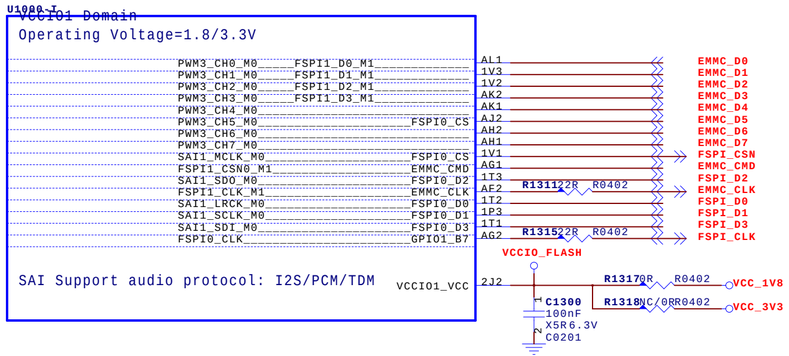
VCC_DDR

C1201 10uF X5R 4V C0402_BGA
C1202 10uF X5R 4V C0402_BGA
C1203 1uF X5R 6.3V C0201
C1204 100nF X5R 6.3V C0201

C1206 10uF X5R 4V C0402_BGA
C1207 100nF X5R 6.3V C0201

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EMMC/FSPI



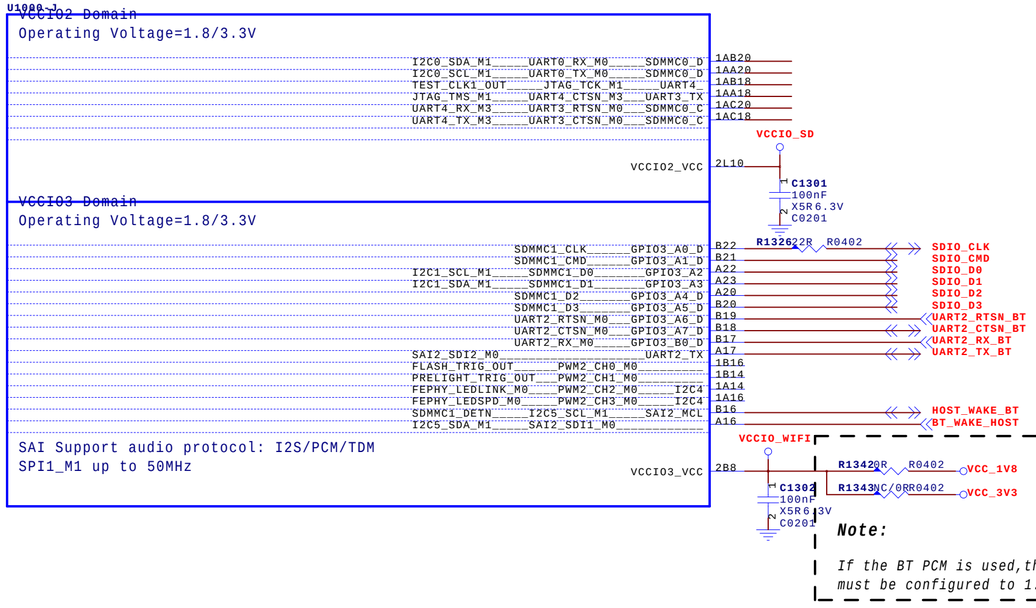
Note:

EMMC design rules:
1. *Max length skew between clk and data < 20ps*
2. *Max trace length < 3 inchs*
3. *Max allowed vias < 2*
4. *Trace impedance 50ohm+/-10%*
5. *The distance between other signals follows the 3W rule.*

Note:

FSPI design rules:
1. *Max length skew between clk and data < 33ps*
2. *Max trace length < 4 inchs*
3. *Trace impedance 50ohm+/-10%*
4. *The distance between other signals follows the 3W rule.*

SDMMC0/SDMMC1/JTAG/UART/I2S2



Note:

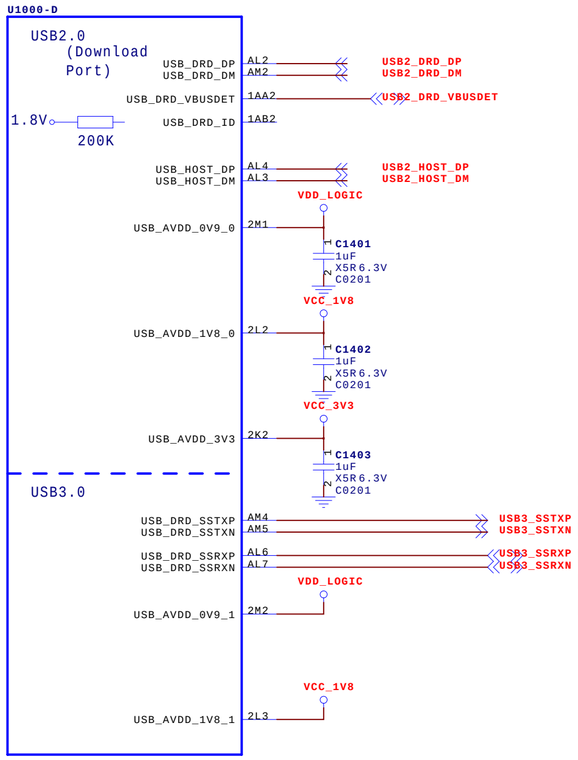
SDMMC design rules:
1. *Max length skew between clk and data < 20ps*
2. *Max trace length < 4 inchs with 50MHz¡UCLK¡U200MHz*
3. *Max trace length < 6 inchs with CLK¡U50MHz*
4. *Max allowed vias < 2*
5. *Trace impedance 50ohm+/-10%*
6. *The distance between other signals follows the 3W rule.*

Note:

If the BT PCM is used,the power domain must be configured to 1.8V

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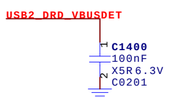
USB Controller



Note:

USB2.0 design rules:

1. ~~Max intra pair skew <3ps~~
2. Max trace length < 6 inchs
3. Max allowed vias < 6
4. Trace impedance 90ohm+/-10%
5. The distance between other signals follows the 3W rule.



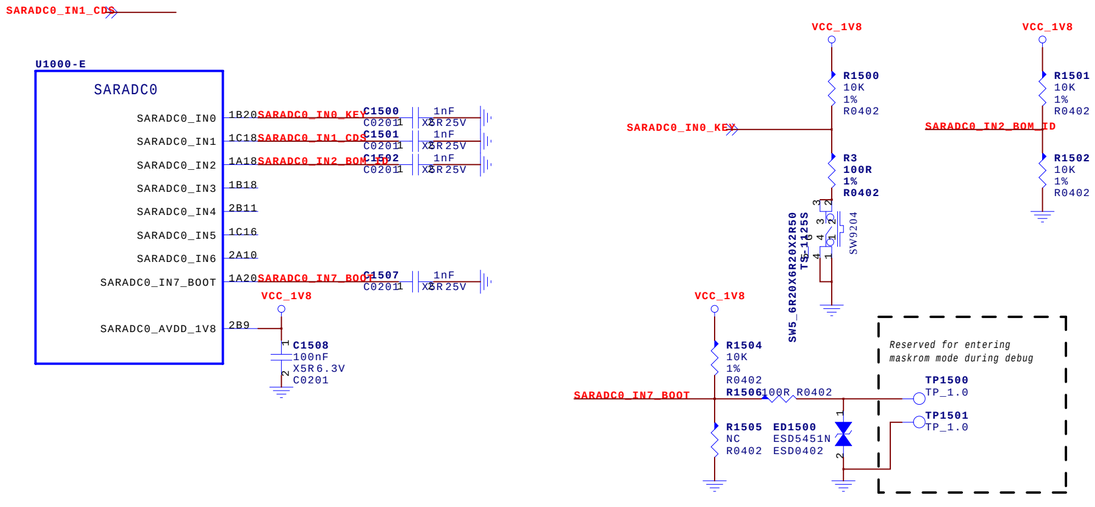
Note:

USB3.0 design rules:

1. ~~Max intra pair skew <1ps~~
2. Max trace length < 6 inchs@Host and 5inchs@Device/DRD
3. Max allowed vias < 2
4. Trace impedance 90ohm+/-10%
5. The distance between other signals follows the 4W rule.

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SARADC0



BOOT Mode Config of SARADC0_IN7

Item	Rup	Rdwon	Boot Mode
Config1	NC	10K	USB/UART0(Maskrom mode)
Config2	100K	12K/27K	FSPI0--SDMMC0--USB/UART0
Config3	100K	51K/82K	FSPI0--EMMC--USB/UART0
Config4	100K	120K	EMMC--USB/UART0
Config5	100K	200K	EMMC--SDMMC0--USB/UART0
Config6	100K	330K/820K	SDMMC0--USB/UART0
Config7	10K	NC	FSPI0--EMMC--SDMMC0--USB/UART0(Auto mode)

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VI_CIF/ETH/SAI/UART/PWM/I2C

U1000-M

VCCIO6 Domain

Operating Voltage=1.8/3.3V

SARADC1_IN0	ETH_PPSTRIG_M0	CAN0_R	1T23
SARADC1_IN1	ETH_PTP_REFCLK_M0	CAN0_T	AD32
SARADC1_IN2	ETH_PPSCLK_M0	CAN1_R	AC32
SARADC1_IN3	ETH_RXD2_M0	CAN1_T	AC31
SARADC1_IN4	ETH_RXD3_M0		AB31
SARADC1_IN5	ETH_TXD2_M0		AA31
SARADC1_IN6	ETH_TXD3_M0		AA32
SARADC1_IN7	ETH_TXD0_M0	SAI0_S	Y32
SARADC2_IN0	ETH_TXD1_M0	SAI0_S	W31
ETH_TXCTL_M0	SAI0_SDO3_M1	VI_CIF	1P21
ETH_RXD0_M0		VI_CIF	1P22
ETH_RXD1_M0		VI_CIF	1P23
ETH_MCLK_M0		VI_CIF	1M21
SARADC2_IN1	ETH_RXCTL_M0		1M22
SARADC2_IN2			1M23
SARADC2_IN3	ETH_MDIO_M0		1K21
SARADC2_IN4	ETH_MDC_M0		V32
SARADC2_IN5	ETH_CLK_25M_OUT_M0		U32
SARADC2_IN6	ETH_TXCLK_M0		U31
SARADC2_IN7	ETH_RXCLK_M0		V31

MOTOR1_D1
MOTOR1_D2
MOTOR1_D3
MOTOR1_D4
MOTOR2_D1
MOTOR2_D2
MOTOR2_D3
MOTOR2_D4
USB_DRD_PWREN_H
USB_PORT_SEL
ETH_LED_ACT
ETH_LED_SPD

SARADC1&2
Input range:0V~3.3V
Full-scale range:0V~1.8V

SAI Support audio protocol: I2S/PCM/TDM

VCCIO6_VCC 2G12

VCC_3V3

C1601 100nF X5R6.3V C0201

I2C/SPI/MIPI-CLK

U1000-K

VCCIO4 Domain

Operating Voltage=1.8/3.3V

CAM_CLK3_OUT	I2C1_SDA_M2	UART4_RT	AG31
CAM_CLK2_OUT	I2C1_SCL_M2	UART4_CT	AG32
PWM0_CH4_M1			1V23
PWM0_CH5_M1	SPI0_CSN1_M1	SAI1_MCL	1V21
SPI0_MOSI_M1	SAI1_SCLK_M1	I2C3_SCL	1V22
SPI0_MISO_M1	SAI1_LRCK_M1	I2C3_SDA	AF32
PWM0_CH6_M1	SPI0_CSN0_M1	SAI1_SDI	1T21
PWM0_CH7_M1	SPI0_CLK_M1	SAI1_SDO	1T22
CAM_CLK1_OUT		UART5_RT	AE31
CAM_CLK0_OUT		UART5_CT	AE31

R1653 22R R0402
R1654 22R R0402

MIPI_RX0_RST
MIPI_RX1_RST
MIPI_RX0_PDN
MIPI_RX1_PDN
I2C3_SCL_CAM
I2C3_SDA_CAM
I2C4_SDA_CAM
I2C4_SCL_CAM
MIPI_MCLK_OUT1
MIPI_MCLK_OUT0

SAI Support audio protocol: I2S/PCM/TDM

VCCIO4_VCC 2J12

VCC_1V8

C1606 100nF X5R6.3V C0201

VCC_1V8

R1617 2.2K R0402
R1618 2.2K R0402
R1656 2.2K R0402
R1657 2.2K R0402

I2C3_SCL_CAM
I2C3_SDA_CAM
I2C4_SCL_CAM
I2C4_SDA_CAM

MIPI-CSI

U1000-F

MIPI DPHY CSI RX0

MIPI_DPHY_CSI_RX0_D0P	LVDS_RX0_D0P	AM25
MIPI_DPHY_CSI_RX0_D0N	LVDS_RX0_D0N	AM26
MIPI_DPHY_CSI_RX0_D1P	LVDS_RX0_D1P	AL24
MIPI_DPHY_CSI_RX0_D1N	LVDS_RX0_D1N	AL25
MIPI_DPHY_CSI_RX0_D2P	LVDS_RX0_D2P	AM19
MIPI_DPHY_CSI_RX0_D2N	LVDS_RX0_D2N	AM20
MIPI_DPHY_CSI_RX0_D3P	LVDS_RX0_D3P	AL20
MIPI_DPHY_CSI_RX0_D3N	LVDS_RX0_D3N	AL21
MIPI_DPHY_CSI_RX0_CLK0P	LVDS_RX0_CLK0P	AM23
MIPI_DPHY_CSI_RX0_CLK0N	LVDS_RX0_CLK0N	AL23
MIPI_DPHY_CSI_RX0_CLK1P	LVDS_RX0_CLK1P	AL22
MIPI_DPHY_CSI_RX0_CLK1N	LVDS_RX0_CLK1N	AM22

MIPI_CSI_RX0_D0P
MIPI_CSI_RX0_D0N
MIPI_CSI_RX0_D1P
MIPI_CSI_RX0_D1N
MIPI_CSI_RX0_D2P
MIPI_CSI_RX0_D2N
MIPI_CSI_RX0_D3P
MIPI_CSI_RX0_D3N
MIPI_CSI_RX0_CLK0P
MIPI_CSI_RX0_CLK0N

VDD_LOGIC

MIPI_DPHY_CSI_RX0_AVDD_0V9 2L12

C1600 100nF X5R6.3V C0201

VCC_1V8

MIPI_DPHY_CSI_RX0_AVDD_1V8 2L11

C1602 100nF X5R6.3V C0201

MIPI DPHY CSI RX1

MIPI_DPHY_CSI_RX1_D0P	LVDS_RX1_D0P	AJ32
MIPI_DPHY_CSI_RX1_D0N	LVDS_RX1_D0N	AJ31
MIPI_DPHY_CSI_RX1_D1P	LVDS_RX1_D1P	AL32
MIPI_DPHY_CSI_RX1_D1N	LVDS_RX1_D1N	AK32
MIPI_DPHY_CSI_RX1_D2P	LVDS_RX1_D2P	AL27
MIPI_DPHY_CSI_RX1_D2N	LVDS_RX1_D2N	AL28
MIPI_DPHY_CSI_RX1_D3P	LVDS_RX1_D3P	AM28
MIPI_DPHY_CSI_RX1_D3N	LVDS_RX1_D3N	AM29
MIPI_DPHY_CSI_RX1_CLK0P	LVDS_RX1_CLK0P	AL31
MIPI_DPHY_CSI_RX1_CLK0N	LVDS_RX1_CLK0N	AM31
MIPI_DPHY_CSI_RX1_CLK1P	LVDS_RX1_CLK1P	AL29
MIPI_DPHY_CSI_RX1_CLK1N	LVDS_RX1_CLK1N	AL30

MIPI_CSI_RX1_D0P
MIPI_CSI_RX1_D0N
MIPI_CSI_RX1_D1P
MIPI_CSI_RX1_D1N
MIPI_CSI_RX1_D2P
MIPI_CSI_RX1_D2N
MIPI_CSI_RX1_D3P
MIPI_CSI_RX1_D3N
MIPI_CSI_RX1_CLK0P
MIPI_CSI_RX1_CLK0N

VDD_LOGIC

MIPI_DPHY_CSI_RX1_AVDD_0V9 2K12

C1604 100nF X5R6.3V C0201

VCC_1V8

MIPI_DPHY_CSI_RX1_AVDD_1V8 2K11

C1605 100nF X5R6.3V C0201

Note:

MIPI CSI RX design rules:

1. *Max intra-pair skew < 2ps*
2. *Max length skew between clk and data < 6ps*
3. *Max trace length < 6 inchs*
4. *Max allowed vias < 4*
5. *Trace impedance 100ohm+/-10%*
6. *The distance between other signals follows the 3W rule.*

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VO_LCDC/VI_CIF/ETH/SAI/UART/PWM/I2C

U1000-J

VCCIO5 Domain

Operating Voltage=1.8/3.3V

	IRFPA_OOC_SDA0	DSMC_CSN1	R31	GMAC_TXD3_M1
	IRFPA_OOC_SDA1	DSMC_DQS1 SAI2_SDI	R32	I2C5_SCL_TP
	DSMC_D15	SAI2_SDI1_M1 ETH_PPSC	P32	LCD_BL_PWM
	DSMC_D14	SAI2_MCLK_M1 SPI0_CSN	P31	LCD_PWREN_H
	DSMC_D13	SAI2_SDO_M1 SPI0_MOS	N31	TP_INT_L
	DSMC_D12	SAI2_SCLK_M1 SPI0_MIS	M32	TP_RST_L
	DSMC_D11	SAI2_SDI0_M1 SPI0_CLK	L32	USB2_HOST_PWREN_H
	I2C5_SDA_M2	DSMC_D10 SAI2_LRC	L31	I2C5_SDA_TP
	IRFPA_OOC_SDA2	DSMC_D9	K31	GMAC_RXCTL_M1
	IRFPA_OOC_SDA3	DSMC_D8	1K23	GMAC_RXD0_M1
	DSMC_RESETN	DSMC_INT1 ETH_RXD1	1H23	GMAC_RXD1_M1
	DSMC_RDYN	ETH_MCLK	2E13	
	DSMC_CSN0		1H22	GEPHY_RST_3V3IO
	DSMC_DQS0	ETH_MDIO	1H21	GMAC_MDIO_M1
	DSMC_CLKN	DSMC_INT0 ETH_MDC_	J31	GMAC_MDC_M1
	DSMC_CLKP	ETH_TXD0	H32	GMAC_TXD0_M1
	IRFPA_OOC_SDA4	DSMC_D7	J32	GMAC_TXD1_M1
	IRFPA_OOC_SDA5	DSMC_D6	1F22	CLK_OUT_ETHERNET_M1
	IRFPA_OOC_SDA6	DSMC_D5	1F23	GMAC_TXCTL_M1
	DSMC_D4	SAI1_MCLK_M2 ETH_RXD2	G31	GMAC_RXD2_M1
	DSMC_D3	SAI1_SDO_M2 ETH_RXD3	F31	GMAC_RXD3_M1
	DSMC_D2	SAI1_SCLK_M2 ETH_TXD2	1D23	GMAC_TXD2_M1
	DSMC_D1	SAI1_LRCK_M2 ETH_TXCL	1D22	GMAC_TXCLK_M1
	DSMC_D0	SAI1_SDI_M2 ETH_RXCL	1D21	GMAC_RXCLK_M1
	I2C3_SCL_M2	DSMC_CSN3	F32	IRCUT_A0
	I2C3_SDA_M2	DSMC_CSN2	E31	IRCUT_B0
	DSMC_INT3		E32	IRCUT_A1
	DSMC_INT2		D31	IRCUT_B1
	I2C2_SCL_M1	FEPHY_LE	2D13	
	I2C2_SDA_M1	FEPHY_LE	1C22	
	IRFPA_FSYNC		1A22	SATA_LED
	IRFPA_MCLK		1B22	WORK_PWM_LED

R1702 22R R0402 R1703 22R R0402 R1704 22R R0402 R1705 22R R0402 R1706 22R R0402 R1707 22R R0402 R1708 22R R0402 R1709 22R R0402

SAI Support audio protocol: I2S/PCM/TDM

VCCIO5_VCC_0 2E12
VCCIO5_VCC_1 2E12

VCC_3V3

C1700 100nF X5R6.3V C0201
C1701 100nF X5R6.3V C0201

VCC_3V3

R1700 4.7K R0402 — I2C5_SCL_TP
R1701 4.7K R0402 — I2C5_SDA_TP

MIPI-DSI Interface

U1000-G

MIPI DPHY DSI TX

MIPI_DPHY_DSI_TX_D0P	B28	MIPI_DSI_D0P
MIPI_DPHY_DSI_TX_D0N	B27	MIPI_DSI_D0N
MIPI_DPHY_DSI_TX_D1P	A29	MIPI_DSI_D1P
MIPI_DPHY_DSI_TX_D1N	A28	MIPI_DSI_D1N
MIPI_DPHY_DSI_TX_D2P	A31	MIPI_DSI_D2P
MIPI_DPHY_DSI_TX_D2N	B31	MIPI_DSI_D2N
MIPI_DPHY_DSI_TX_D3P	C32	MIPI_DSI_D3P
MIPI_DPHY_DSI_TX_D3N	B32	MIPI_DSI_D3N
MIPI_DPHY_DSI_TX_CLKP	B30	MIPI_DSI_CLKP
MIPI_DPHY_DSI_TX_CLKN	B29	MIPI_DSI_CLKN

VDD_LOGIC

MIPI_DSI_TX0_AVDD_0V9 2B12 — C1702 100nF X5R6.3V C0201

VCC_1V8

MIPI_DSI_TX0_AVDD_1V8 2C12 — C1703 100nF X5R6.3V C0201

Note:

MIPI DSI RX design rules:

1. *Max intra-pair skew < 2ps*
2. *Max length skew between clk and data < 6ps*
3. *Max trace length < 6 inchs*
4. *Max allowed vias < 4*
5. *Trace impedance 100ohm+/-10%*
6. *The distance between other signals follows the 3W rule.*

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Audio Interface

U1000-N

VCCIO7 Domain

Operating Voltage=1.8/3.3V

SAI0_SCLK_M0	1Y21	SAI0_SCLK	
PDM_CLK1_M0	1AA22	I2C4_SCL_M3	
SAI0_MCLK_M0	1AA16		
SAI0_LRCK_M0	1AB22	SAI0_LRCK	
PDM_CLK0_M0	1AC16	I2C4_SDA_M3	
SAI0_SDO0_M0	1AC22	SAI0_SDO	
SAI0_SDI0_M0	1AB16	PDM_SDI0	SAI0_SDI
SAI0_SDO1_M0	1AA14	SAI0_SDI3_M0	PDM_SDI3
SAI0_SDO2_M0	1AB14	SAI0_SDI2_M0	PDM_SDI2 DSM_AUD_RN
SAI0_SDO3_M0	1AC14	SAI0_SDI1_M0	PDM_SDI1 DSM_AUD_RP

SAI Support audio protocol: I2S/PCM/TDM

VCCIO7_VCC

2L9 — C1800 100nF X5R6.3V C0201

R1800 NC/0R R0402 — VCC_1V8

R1801 0R R0402 — VCC_3V3

Note:

If the BT PCM is used,the power domain must be configured to 1.8V

Audio ADC

U1000-O

AUDIO ADC0(PMU)

PMU_MICP_AUDIO_ADC0_MICP	AM16	AUDIO_ADC0_MICP	C1801 C0402	1uF X5R6.3V	MIC0_P
PMU_MICN_AUDIO_ADC0_MICN	AM17	AUDIO_ADC0_MICN	C1802 C0402	1uF X5R6.3V	MIC0_N
AUDIO_ADC0_VCM	2M6	AUDIO_ADC0_VCM	C1803 C0201	2.2uF X5R4V	
AUDIO_ADC0_VREF	2M7	AUDIO_ADC0_VREF	C1804 C0201	2.2uF X5R4V	

AUDIO ADC1

AUDIO_ADC1_MICP	AL17	AUDIO_ADC1_MICP	C1805 C0402	1uF X5R6.3V	MIC1_P
AUDIO_ADC1_MICN	AL18	AUDIO_ADC1_MICN	C1806 C0402	1uF X5R6.3V	MIC1_N
AUDIO_ADC1_VCM	2L6	AUDIO_ADC1_VCM	C1807 C0201	2.2uF X5R4V	
AUDIO_ADC1_VREF	2N8	AUDIO_ADC1_VREF	C1808 C0201	2.2uF X5R4V	

AUDIO ADC Power

AUDIO_ADC_AVDD_1V8 — 2N7 — AVDD1V8_AUDIO

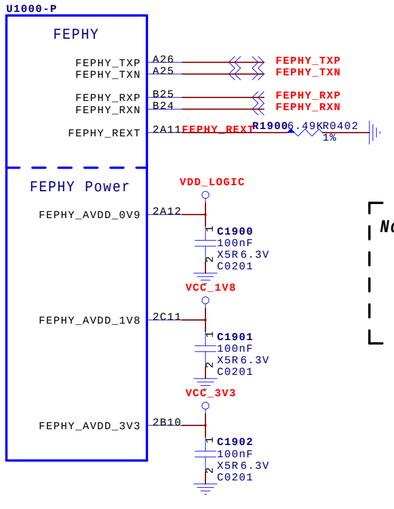
R1815 0R R0402 — VCC_1V8

C1809 4.7uF X5R6.3V C0402

C1810 100nF X5R6.3V C0201

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FEPHY



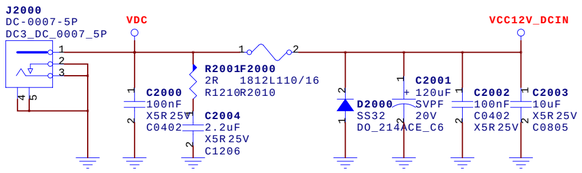
Note:

FEPHY design rules:

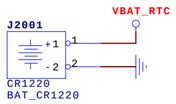
1. ~~*Max intra pair skew < 3ps*~~
2. *Max trace length < 4 inchs*
3. *Max allowed vias < 2*
4. *Trace impedance 100ohm+/-10%*
5. *The distance between other signals follows the 3W rule.*

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12V DCIN
Setp 0



RTC Power

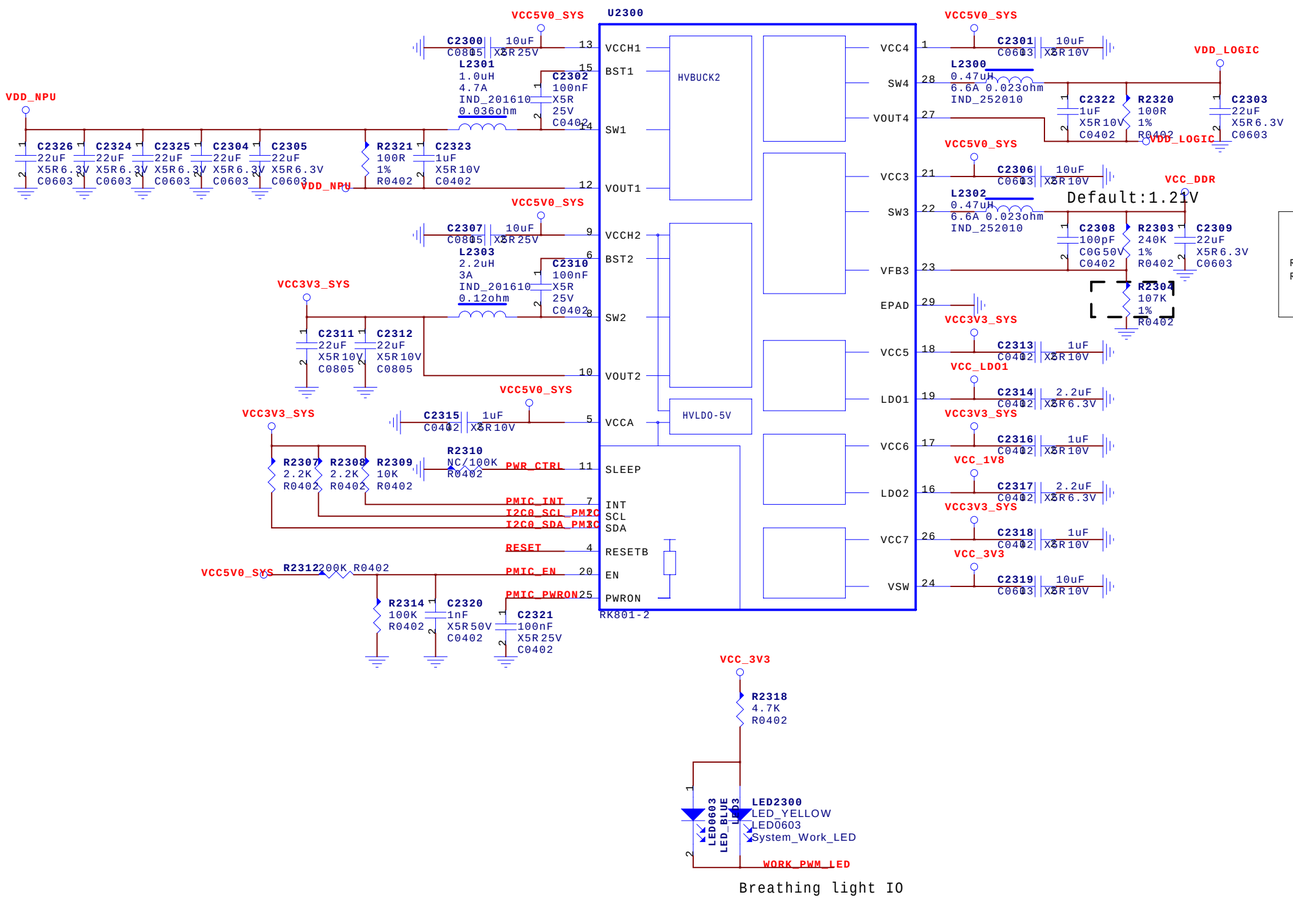


12V--->VCC5V0_SYS
Setp 1

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PMIC RK801_5V(opt)

I2C0_SCL_PMIC
I2C0_SDA_PMIC
RESET
PMIC_INT
PWR_CTRL
WORK_PWM_LED

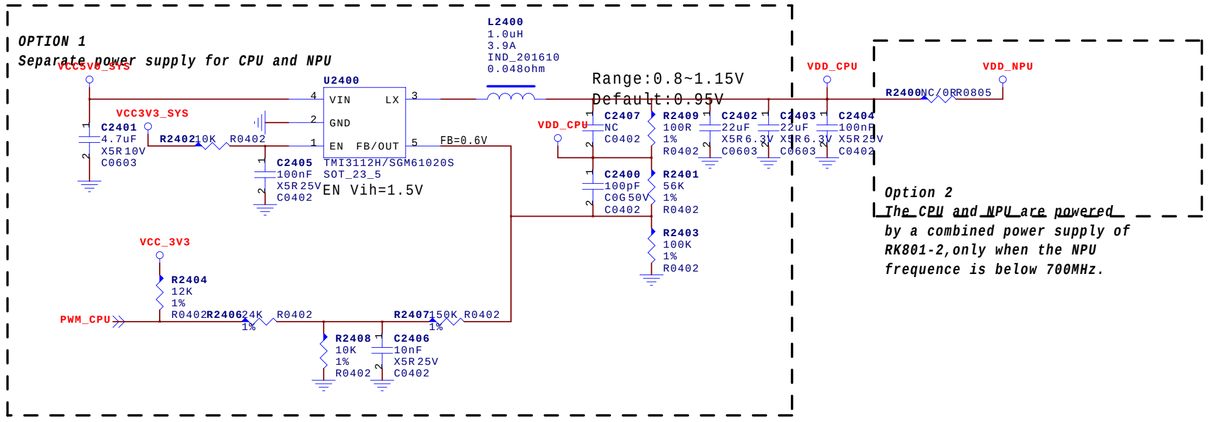


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VCC5V0_SYS--->VDD_CPU

Step 2



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USB2.0

USB2_DRD_DP
USB2_DRD_DM
Diff 90 Ohm ¡Ã10%
USB2_DRD_VBUSDET

Type-C Device

HUB_DM2 — R2500 0R R0402
HUB_DP2 — R2503 0R R0402
ED2500 ESD5341N ESD0402
ED2501 ESD5341N ESD0402
VBUS5V0_HOST
J3: 5V, D0-, D0+, GND, 5V, D1-, D1+, GND, GND GND GND GND
HUB_D2M
HUB_D2P
2XUSB2.0
HUB_DM3 — R5 0R R0402 — HUB_D3M
HUB_DP3 — R4 0R R0402 — HUB_D3+
ED1 ESD5341N ESD0402
ED2 ESD5341N ESD0402
VCC5V0_SYS
R2506 10K R0402
R2507 15K R0402

C158 NC/27pF
Y6 12MHz_CL20pF_20ppm XT-SMD4P3225
12M_X1
12M_X2
C119 NC/27pF

PL2586 (FE1.1S)
GND, XTAL1, XTAL2, DM1, DP1, DM2, DP2, DM3, DP3, DM4, DP4, AVDD, VDD, REXT
VD18, TESTJ, OVCJ, PWRJ, LED2, LED1, DRV, VD33, VDD5, BUSJ, VBUSM, XRSTJ, DPU, DMU

6C23 100NF C0402 — VDD12
R115 0R/NC — HUB_VDD
C188 10uF_6.3V — VCC5V0_SYS
C204 10uF_10V
R107 10K
R106 10K
R105 10K
R164 22K — HUB_VDD
C207 1uF_6.3V
USB2_DRD_D+
USB2_DRD_D-
C172 100N — R108 0R — VDD12
HUB_VDD
6C13 100NF C0402
R114 330R_1%

J9 NC/4P/2.0
USB2_DRD_D-
USB2_DRD_D+
VCC5V0_SYS

R1 0R R0402 — USB2_DRD_D+
R2 0R R0402 — USB2_DRD_D-

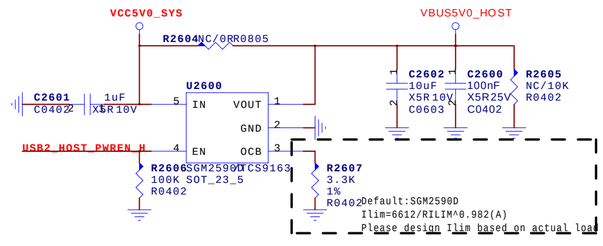
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USB2.0 Type-A HOST

USB2_HOST_DP
USB2_HOST_DM
Diff 90 Ohm ¡Ä10%

USB2_HOST_PWREN_H

USB2.0 POWER



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LPDDR3 1x32bit

LPDDR3_D0 LPDDR3_D1 LPDDR3_D2 LPDDR3_D3 LPDDR3_D4 LPDDR3_D5 LPDDR3_D6 LPDDR3_D7 LPDDR3_DM0 LPDDR3_DQS0P LPDDR3_DQS0N
LPDDR3_D8 LPDDR3_D9 LPDDR3_D10 LPDDR3_D11 LPDDR3_D12 LPDDR3_D13 LPDDR3_D14 LPDDR3_D15 LPDDR3_DM1 LPDDR3_DQS1P LPDDR3_DQS1N
LPDDR3_A0 LPDDR3_A1 LPDDR3_A2 LPDDR3_A3 LPDDR3_A4 LPDDR3_A5 LPDDR3_A6 LPDDR3_A7 LPDDR3_A8 LPDDR3_A9
LPDDR3_D16 LPDDR3_D17 LPDDR3_D18 LPDDR3_D19 LPDDR3_D20 LPDDR3_D21 LPDDR3_D22 LPDDR3_D23 LPDDR3_DM2 LPDDR3_DQS2P LPDDR3_DQS2N
LPDDR3_D24 LPDDR3_D25 LPDDR3_D26 LPDDR3_D27 LPDDR3_D28 LPDDR3_D29 LPDDR3_D30 LPDDR3_D31 LPDDR3_DM3 LPDDR3_DQS3P LPDDR3_DQS3N
LPDDR3_CKE LPDDR3_CS0N LPDDR3_CS1N LPDDR3_ODT0 LPDDR3_CLKP LPDDR3_CLKN

LPDDR3_CLKP R3102 49.9R R0402 1%
LPDDR3_CLKN R3103 49.9R R0402 1%
C3100 NC/100pF C0402

U3100-A

LPDDR3_D1	P9	DQ0	CA0	R2	LPDDR3_A0
LPDDR3_D6	N9	DQ1	CA1	P2	LPDDR3_A1
LPDDR3_D3	N10	DQ2	CA2	N2	LPDDR3_A2
LPDDR3_D7	N11	DQ3	CA3	N3	LPDDR3_A3
LPDDR3_D5	M8	DQ4	CA4	M3	LPDDR3_A4
LPDDR3_D0	M9	DQ5	CA5	F3	LPDDR3_A5
LPDDR3_D4	M10	DQ6	CA6	E3	LPDDR3_A6
LPDDR3_D2	M11	DQ7	CA7	F2	LPDDR3_A7
			CA8	D2	LPDDR3_A8
LPDDR3_DQS0P	L10	DQS0P	CA9	C2	LPDDR3_A9
LPDDR3_DQS0N	L11	DQS0N			
LPDDR3_DM0	L8	DM0	CLKP	J3	LPDDR3_CLKP
			CLKN	J2	LPDDR3_CLKN
LPDDR3_D9	F11	DQ8	CKE0	K3	LPDDR3_CKE
LPDDR3_D8	F10	DQ9	CKE1	K4	
LPDDR3_D15	F9	DQ10			
LPDDR3_D14	F8	DQ11	CS0	L3	LPDDR3_CS0N
LPDDR3_D11	E11	DQ12	CS1	L4	LPDDR3_CS1N
LPDDR3_D10	E10	DQ13			
LPDDR3_D13	E9	DQ14	ODT	J8	LPDDR3_ODT0
LPDDR3_D12	D9	DQ15			
LPDDR3_DQS1P	G10	DQS1P			
LPDDR3_DQS1N	G11	DQS1N			
LPDDR3_DM1	G8	DM1			
LPDDR3_D16	T8	DQ16	ZQ0	B3	R3100 240R R0402 1%
LPDDR3_D20	T9	DQ17			
LPDDR3_D19	T10	DQ18			
LPDDR3_D18	T11	DQ19			
LPDDR3_D21	R8	DQ20			
LPDDR3_D17	R9	DQ21			
LPDDR3_D23	R10	DQ22	ZQ1	B4	R3101 240R R0402 1%
LPDDR3_D22	R11	DQ23			
LPDDR3_DQS2P	P10	DQS2P			
LPDDR3_DQS2N	P11	DQS2N			
LPDDR3_DM2	P8	DM2	NU1	A1	
			NU2	A2	
			NU3	A12	
LPDDR3_D24	C11	DQ24	NU4	A13	
LPDDR3_D25	C10	DQ25	NU5	B1	
LPDDR3_D26	C9	DQ26	NU6	B13	
LPDDR3_D27	C8	DQ27	NU7	T1	
LPDDR3_D30	B11	DQ28	NU8	T13	
LPDDR3_D31	B10	DQ29	NU9	U1	
LPDDR3_D28	B9	DQ30		U2	
LPDDR3_D29	B8	DQ31	NU10	U12	
			NU11	U13	
LPDDR3_DQS3P	D10	DQS3P	NU12		
LPDDR3_DQS3N	D11	DQS3N			
			NC1	C4	
LPDDR3_DM3	D8	DM3	NC2	K9	
			NC3	R3	
VREF_LP3_DQ	J11	VREFDQ	VREFCA	H4	VREF_LP3_CA

VDD1

U3100-B

VDD1_1V8_DDR	A3	VDD1_1	VSS1	B2
	A4	VDD1_2	VSS2	B5
	A5	VDD1_3	VSS3	C5
	A6	VDD1_4	VSS4	E4
	A10	VDD1_5	VSS5	E5
	U3	VDD1_6	VSS6	F5
	U4	VDD1_7	VSS7	H2
	U5	VDD1_8	VSS8	J12
	U6	VDD1_9	VSS9	K2
	U10	VDD1_10	VSS10	L6
			VSS11	M5
VCC_DDR	U9	VDD2_1	VSS12	N4
	U8	VDD2_2	VSS13	N5
	P6	VDD2_3	VSS14	R4
	P5	VDD2_4	VSS15	R5
	P4	VDD2_5	VSS16	T2
	L5	VDD2_6	VSS17	T3
	K12	VDD2_7	VSS18	T4
	K6	VDD2_8	VSS19	T5
	K5	VDD2_9		
	J6	VDD2_10		
	J5	VDD2_11		
	H12	VDD2_12		
	H6	VDD2_13		
	H5	VDD2_14		
	G5	VDD2_15		
	D6	VDD2_16		
	D5	VDD2_17		
	D4	VDD2_18		
	A9	VDD2_19	VSSCA1	C3
	A8	VDD2_20	VSSCA2	D3
			VSSCA3	F4
	M2	VDDCA1	VSSCA4	G3
	L2	VDDCA2	VSSCA5	G4
	H3	VDDCA3	VSSCA6	J4
	G2	VDDCA4	VSSCA7	M4
	F2	VDDCA5	VSSCA8	P3
VCC_DDR	U11	VDDQ_1	VSSQ1	B6
	R12	VDDQ_2	VSSQ2	B12
	N12	VDDQ_3	VSSQ3	C6
	N8	VDDQ_4	VSSQ4	D12
	L12	VDDQ_5	VSSQ5	F6
	K11	VDDQ_6	VSSQ6	F6
	K8	VDDQ_7	VSSQ7	F12
	J10	VDDQ_8	VSSQ8	G6
	J9	VDDQ_9	VSSQ9	G9
	H11	VDDQ_10	VSSQ10	H10
	H9	VDDQ_11	VSSQ11	K10
	H8	VDDQ_12	VSSQ12	L9
	G12	VDDQ_13	VSSQ13	M6
	E12	VDDQ_14	VSSQ14	M12
	E8	VDDQ_15	VSSQ15	N6
	C12	VDDQ_16	VSSQ16	P12
	A11	VDDQ_17	VSSQ17	R6
			VSSQ18	T6
			VSSQ19	T12

LP3S_B178
FBGA178_13R00X11R50X1R20

Note:All the power filter capacitors should be placed close to the power pins of LPDDR3

VCC_DDR
C3101 10uF X5R10V C0603
C3102 1uF X5R6.3V C0402
C3103 1uF X5R6.3V C0402
C3108 1uF X5R6.3V C0402
C3105 1uF X5R6.3V C0402
C3106 100nF X5R25V C0402
C3104 100nF X5R25V C0402

VCC_1V8 R3105 0R R0402 VDD1_1V8_DDR
C3109 10uF X5R10V C0603
C3111 100nF X5R25V C0402
C3112 100nF X5R25V C0402

VCC_DDR
C3120 10uF X5R10V C0603
C3121 1uF X5R6.3V C0402
C3129 1uF X5R6.3V C0402
C3132 1uF X5R6.3V C0402
C3125 100nF X5R25V C0402
C3122 100nF X5R6.3V C0201
C3130 100nF X5R6.3V C0201
C3131 100nF X5R6.3V C0201

VCC_DDR
R3104 10K 1% R0402
C3114 100nF X5R25V C0402
R3106 10K 1% R0402
C3116 100nF X5R25V C0402

Put Together

VREF_LP3_CA
C3117 1nF X5R50V C0402

Close to pin H4

VREFOUT_DDR R3108 0R R0402 VREF_LP3_DQ
C3118 1nF X5R50V C0402

Close to pin J11

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eMMC

EMMC_D0
EMMC_D1
EMMC_D2
EMMC_D3
EMMC_D4
EMMC_D5
EMMC_D6
EMMC_D7
EMMC_CMD
EMMC_CLK

VCCIO_FLASH

R4000 10K R0402
R4001 10K R0402

EMMC_D0
EMMC_D1
EMMC_D2
EMMC_D3
EMMC_D4
EMMC_D5
EMMC_D6
EMMC_D7
EMMC_CMD
EMMC_CLK

C4006 NC C0402

VCCIO_FLASH

R4004 4.7K R0402

EMMC_RST

C4005 2.2uF C0402 X5R6.3V

U4000-A

A3	DATA0	VCCQ1	C6
A4	DATA1	VCCQ2	M4
A5	DATA2	VCCQ3	N4
B2	DATA3	VCCQ4	P3
B3	DATA4	VCCQ5	P5
B4	DATA5		
B5	DATA6		
B6	DATA7		
M5	CMD		
M6	CLK	VCC1	E6
		VCC2	F5
		VCC3	J10
		VCC4	K9
H5	DATA_STROBE	VSS1	J5
		VSS2	A6
		VSS3	C4
		VSS4	E7
		VSS5	G5
		VSS6	H10
		VSSQ1	K8
		VSSQ2	N2
		VSSQ3	N5
		VSSQ4	P4
		VSSQ5	P6
K5	RST_N		
		VSF4	K10
		VSF3	F10
C2	VDDI	VSF2	E10
		VSF1	E9

VCCIO_FLASH

C4000 4.7uF X5R6.3V C0402
C4001 100nF X5R25V C0402
C4002 100nF X5R25V C0402

VCC_3V3

C4003 4.7uF X5R6.3V C0402
C4004 100nF X5R25V C0402

U4000-B

A2	NC2	NC196	P14
A8	NC8	NC195	P13
A9	NC9	NC194	P12
A10	NC10	NC193	P11
A11	NC11	NC191	P9
A12	NC12	NC190	P8
A13	NC13	NC184	P2
A14	NC14	NC183	P1
B1	NC15	NC182	N14
B7	NC21	NC181	N13
B8	NC22	NC180	N12
B9	NC23	NC179	N11
B10	NC24	NC178	N10
B11	NC25	NC177	N9
B12	NC26	NC176	N8
B13	NC27	NC175	N7
B14	NC28	NC174	N6
C1	NC29	NC171	N3
C3	NC31	NC169	N1
C7	NC35	NC168	M14
C8	NC36	NC167	M13
C9	NC37	NC166	M12
C10	NC38	NC165	M11
C11	NC39	NC164	M10
C12	NC40	NC163	M9
C13	NC41	NC162	M8
C14	NC42	NC161	M7
D1	NC43	NC157	M3
D2	NC44	NC156	M2
D3	NC45	NC155	M1
D4	NC46	NC154	L14
D12	NC54	NC153	L13
D13	NC55	NC152	L12
D14	NC56	NC143	L3
E1	NC57	NC142	L2
E2	NC58	NC141	L1
E3	NC59	NC140	K14
E12	NC68	NC139	K13
E13	NC69	NC138	K12
E14	NC70	NC129	K3
F1	NC71	NC128	K2
F2	NC72	NC127	K1
F3	NC73	NC126	J14
F12	NC82	NC125	J13
F14	NC84	NC124	J12
G1	NC85	NC115	J3
G2	NC86	NC114	J2
G12	NC96	NC113	J1
G13	NC97	NC112	H14
G14	NC98	NC111	H13
		NC110	H12
		NC101	H3
		NC100	H2
		NC99	H1
A7	RFU1		
E5	RFU2	RFU9	P10
E8	RFU3	RFU8	P7
G3	RFU4	RFU7	K7
G10	RFU5	RFU6	K6

EMMC
BGA153_13RX11R5X0R9_2L

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MIPI CAM0 Interfae

MIPI_CSI_RX0_D0N
MIPI_CSI_RX0_D0P
MIPI_CSI_RX0_D1N
MIPI_CSI_RX0_D1P
MIPI_CSI_RX0_D2N
MIPI_CSI_RX0_D2P
MIPI_CSI_RX0_D3N
MIPI_CSI_RX0_D3P
MIPI_CSI_RX0_CLK0N
MIPI_CSI_RX0_CLK0P

MIPI_MCLK_OUT0

MIPI_RX0_PDN

MIPI_RX0_RST

I2C3_SCL_CAM
I2C3_SDA_CAM

J4700

MIPI RX/LVDS RX/HDMI RX to MIPI

Horizontal SMT
0.5mm Pitch
40Pin
Upper contact

1	GND1		
MIPI_CSI_RX0_D0N	2	MIPI_RX_D0N	
MIPI_CSI_RX0_D0P	3	MIPI_RX_D0P	
4	GND2		
MIPI_CSI_RX0_D1N	5	MIPI_RX_D1N	
MIPI_CSI_RX0_D1P	6	MIPI_RX_D1P	
7	GND3		
MIPI_CSI_RX0_CLK0N	8	MIPI_RX_CLK0N	
MIPI_CSI_RX0_CLK0P	9	MIPI_RX_CLK0P	
10	GND4		
MIPI_CSI_RX0_D2N	11	MIPI_RX_D2N	
MIPI_CSI_RX0_D2P	12	MIPI_RX_D2P	
13	GND5		1
MIPI_CSI_RX0_D3N	14	MIPI_RX_D3N	
MIPI_CSI_RX0_D3P	15	MIPI_RX_D3P	
16	GND6		
MIPI_MCLK_OUT0	17	MIPI_RX_MCLK0	
18	GND7		
19	MIPI_RX_PDN1/HDMI_RX_INT		
MIPI_RX0_RST	20	MIPI_RX_RST0	
MIPI_RX0_PDN	21	MIPI_RX_PDN0	
22	MIPI_RX_PWREN/1V8_OUT		
I2C3_SCL_CAM	23	MIPI_RX_I2C_SCL/SPI_CLK	
I2C3_SDA_CAM	24	MIPI_RX_I2C_SDA/SPI_MOSI	
25	MIPI_RX_XHS/SPI_MISO/VSYNC/HOTPLUG_DET		
26	MIPI_RX_XVS/SPI_CSN0/FSIN/HDMI_RX_PWM		
27	GND9		40
VCC_MIPI	28	5V0_1	
29	5V0_2		
30	5V0_3		
31	GND10		
32	IRCUT_A		
33	IRCUT_B		
34	MIPI_RX_RST1/HDMI_RX_RST		
35	GND11		
36	MIPI_RX_MCLK1/HDMI_RX_MCLK		
37	GND12		
38	MIPI_RX_CLK1N		
39	MIPI_RX_CLK1P		
40	GND13		

GND_PAD1 41
GND_PAD2 42

Video_FPC_MIPI
CN40FR_0R50_H

MIPI_MCLK_OUT0 — **C4700** C0402 1 — NC 2

VCC_3V3 — **R4704** 0R R0603 — VCC_MIPI

C4702 10uF X5R10V C0603 1 2 — **C4703** 100nF X5R25V C0402 1 2

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MIPI CAM1 Interfae

MIPI_CSI_RX1_D0N
MIPI_CSI_RX1_D0P
MIPI_CSI_RX1_D1N
MIPI_CSI_RX1_D1P
MIPI_CSI_RX1_D2N
MIPI_CSI_RX1_D2P
MIPI_CSI_RX1_D3N
MIPI_CSI_RX1_D3P
MIPI_CSI_RX1_CLK0N
MIPI_CSI_RX1_CLK0P

MIPI_MCLK_OUT1

MIPI_RX1_PDN

MIPI_RX1_RST

I2C4_SCL_CAM
I2C4_SDA_CAM

J4800
MIPI RX/LVDS RX/HDMI RX to MIPI
Horizontal SMT
0.5mm Pitch
40PIN
Upper contact

	1	GND1	
MIPI_CSI_RX1_D0N	2	MIPI_RX_D0N	
MIPI_CSI_RX1_D0P	3	MIPI_RX_D0P	
	4	GND2	
MIPI_CSI_RX1_D1N	5	MIPI_RX_D1N	
MIPI_CSI_RX1_D1P	6	MIPI_RX_D1P	
	7	GND3	
MIPI_CSI_RX1_CLK0N	8	MIPI_RX_CLK0N	
MIPI_CSI_RX1_CLK0P	9	MIPI_RX_CLK0P	
	10	GND4	
MIPI_CSI_RX1_D2N	11	MIPI_RX_D2N	
MIPI_CSI_RX1_D2P	12	MIPI_RX_D2P	
	13	GND5	1
MIPI_CSI_RX1_D3N	14	MIPI_RX_D3N	
MIPI_CSI_RX1_D3P	15	MIPI_RX_D3P	
	16	GND6	
MIPI_MCLK_OUT1	17	MIPI_RX_MCLK0	
	18	GND7	
	19	MIPI_RX_PDN1/HDMI_RX_INT	
MIPI_RX1_RST	20	MIPI_RX_RST0	
MIPI_RX1_PDN	21	MIPI_RX_PDN0	
	22	MIPI_RX_PWREN/1V8_OUT	
I2C4_SCL_CAM	23	MIPI_RX_I2C_SCL/SPI_CLK	
I2C4_SDA_CAM	24	MIPI_RX_I2C_SDA/SPI_MOSI	
	25	MIPI_RX_XHS/SPI_MISO/VSYNC/HOTPLUG_DET	
	26	MIPI_RX_XVS/SPI_CSN0/FSIN/HDMI_RX_PWM	
	27	GND9	40
VCC_MIPI	28	5V0_1	
	29	5V0_2	
	30	5V0_3	
	31	GND10	
	32	IRCUT_A	
	33	IRCUT_B	
	34	MIPI_RX_RST1/HDMI_RX_RST	
	35	GND11	
	36	MIPI_RX_MCLK1/HDMI_RX_MCLK	
	37	GND12	
	38	MIPI_RX_CLK1N	
	39	MIPI_RX_CLK1P	
	40	GND13	
		GND_PAD1	41
		GND_PAD2	42

Video_FPC_MIPI
CN40FR_0R50_H

MIPI_MCLK_OUT1 — **C4800** C0402 — 1 — NC — 2

VCC_MIPI

C4802 10uF X5R 10V C0603 — 1 — 2
C4803 100nF X5R 25V C0402 — 1 — 2

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MIPI-DSI Interface

MIPI_DSI_D0N
MIPI_DSI_D0P
MIPI_DSI_D1N
MIPI_DSI_D1P
MIPI_DSI_D2N
MIPI_DSI_D2P
MIPI_DSI_D3N
MIPI_DSI_D3P
MIPI_DSI_CLKN
MIPI_DSI_CLKP

LCD_BL_PWM
LCD_PWREN_H
TP_INT_L
TP_RST_L
I2C5_SCL_TP
I2C5_SDA_TP

J5000
MIPI_DSI_TX

	1	GND1	
MIPI_DSI_D0N	2	MIPI_D0N	
MIPI_DSI_D0P	3	MIPI_D0P	
	4	GND2	
MIPI_DSI_D1N	5	MIPI_D1N	
MIPI_DSI_D1P	6	MIPI_D1P	
	7	GND3	
MIPI_DSI_CLKN	8	MIPI_CLKN	
MIPI_DSI_CLKP	9	MIPI_CLKP	
	10	GND4	
MIPI_DSI_D2N	11	MIPI_D2N	
MIPI_DSI_D2P	12	MIPI_D2P	
	13	GND5	
MIPI_DSI_D3N	14	MIPI_D3N	
MIPI_DSI_D3P	15	MIPI_D3P	
	16	GND6	
LCD_BL_PWM	17	LCD_PWM_BL	
	18	LCD_RST	
TP5000	19	3V3	
TP5001	20	LCD_EN	
TP5002	21	LCD_CS	
LCD_PWREN_H	22	BL_EN	
I2C5_SCL_TP	23	I2C_SCL	
I2C5_SDA_TP	24	I2C_SDA	
TP_INT_L	25	TP_INT	
TP_RST_L	26	TP_RST	
	27	GND7	
	28	5V0_1	
	29	5V0_2	
VCC5V0_SYS	30	5V0_3	
	31	PAD1	
	32	PAD2	

MainBoard

MIPI_DSI_TXÐÀºÂÊ¹ÓÃ´ÊÁ¬½Ó×ù
ÌåÀàÐÍ£ºFPC
30PIN
ÊÎ¼Ó´¥

1
30

FPC_DSI_MB
CNN30_1R00_FP05SL_V

VCC5V0_SYS

C5000
10uF
10V X5R
C0603

C5001
100nF
25V X5R
C0402

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WIFI/BT Module

SDIO_CLK
SDIO_CMD
SDIO_D0
SDIO_D1
SDIO_D2
SDIO_D3
UART2_RTSN_BT
UART2_CTSN_BT
UART2_RX_BT
UART2_TX_BT
WIFI_WAKE_HOST_3V3
WIFI_REG_ON_3V3
HOST_WAKE_BT
BT_WAKE_HOST
HOST_WAKE_BT
WIFI_PWREN_L

VCCIO_WIFI
C6008 1uF X5R6.3V C0402
C6009 100nF X5R25V C0402

Reserve for 7668 & 8822bs BT_ANT

RF Microstrip
Z0= 50 ohm
C11 10pF 50V C0G C0402
C12 NC
C13 NC

RF Microstrip
Z0= 50 ohm
C6000 10pF C0G 50V C0402
C6001 NC
C6002 NC

7J1 FEED GS2 G1 J-MS-156
7J5 FEED GS2 G1 J-MS-156

NC
BT
GND
7J2
BT_ANTB
7C8 0R
7C9 NC
7C46 NC
GND GND GND

U6200
RTL8822BS
AP_WIFI2X2_13X15_L

12	NC/NC/PCIE_PREST_L	BT_HOST_WAKE	50
13	XTAL_OUT	BT_WAKE	49
14	XTAL_IN	GPIO8_9/NC_BT_ANT/NC	48
15	WL_REG_ON	NC/GND/PCIE_TDP	47
16	WL_HOST_WAKE	GPIO4/NC/PCIE_TDN	46
17	SDIO_DATA_CMD	GPIO5/WL_UART_RX/PCIE_RDP	45
18	SDIO_DATA_CLK	NC/WL_UART_TX/PCIE_RDN	44
19	SDIO_DATA_3	UART_CTS_N	43
20	SDIO_DATA_2	UART_RTS_N	42
21	SDIO_DATA_0/HUSB_DP	UART_RXD	41
22	SDIO_DATA_1/HUSB_DM	UART_TXD	40
23	GND_10	GND_12	39
24	NC/SDIO_VSEL/PCIE_PME_L	BT_REG_ON	38
25	VIN_LDO	NC/NC/PCIE_CLKREQ_L	37

WIFI_REG_ON
WIFI_WAKE_HOST
SDIO_D2
SDIO_D3
SDIO_CMD
SDIO_CLK
SDIO_D0
SDIO_D1
SDIO_VSEL

BT_HOST_WAKE
BT_WAKE
HOST_WAKE_BT
WIFI_REG_ON
UART2_RTSN_BT
UART2_CTSN_BT
UART2_TX_BT
UART2_RX_BT

R6000 NC/0R R0402
R6001 NC/0R R0402
Y6000 X1 GND X2 GND NC/37.4MHz CRY4_3R20X2R50X0R80
C6003 NC/18pF C0G 50V C0402
C6004 NC/18pF C0G 50V C0402
37.4MHz: +-10ppm

OPTION 1

C6005 10uF X5R10V C0603
C6006 100nF X5R25V C0402
C6011 NC C0402

VCCIO_WIFI
R6210 0R
R6212 0R

OPTION 2
SDIO voltage select:
0: 3.3V 1:1.8V

L6000 NC/3.3uH IND_3030-15 1.32A 0.08ohm
C6007 NC/4.7uF C0603

NOTE:
up to 500mA

VIN_LDO_OUT
PCM_SYNC
PCM_IN
PCM_OUT
PCM_CLK
LPO
GND_11
VCCIO_WL
VBAT

SAI0_LRCK
SAI0_SDI
SAI0_SDO
SAI0_SCLK

VBAT_WL
VCCIO_WIFI
0.1uF C6213

R0402 NC/4.7K R9805
OPTION 5

OPTION 3
CLK_32K_OUT
R6007 22R R0402
R6008 NC R0402
WIFI_BT_32KIN
C0402 C6010 NC

R6008 0R R0402
R6002 NC/10K R0402
OPTION 4

Power Control for other Modules
VCC3V3_SYS
R6100 0R R0603
VBAT_WL

Level Shift, default for 3.3V
WIFI_REG_ON_3V3
R6009 0R R0402
R6010 NC R0402
WIFI_REG_ON
OPTION for 1.8V

Note:
Yes: option circuit be mounted
No: option circuit not be mounted

TYPE	WIFI a	WIFI b/g/n	WIFI ac	WIFI 5GHz	BT	Crystals	VCCIO_SDIO	OPTION 1	OPTION 2	OPTION 3	OPTION 4	OPTION 5
RK960	No	Yes	No	No	BT5.3	Integrated	1.8V/3.3V	No	No	No	No	No
Module RK960-V12 RTL8189ETV	No	Yes	No	No	No	Integrated	1.8-3.3V	No	No	No	No	No
Module F89FTSM12-W3 RTL8723DS	No	Yes	No	No	BT4.2	Integrated	1.62-3.6V	No	No	No	Yes	Yes
Module 6223A-SRD RTL8821CS-CG	Yes	Yes	Yes	Yes	BLE4.2	Integrated	1.7-3.6V	No	No	No	No	No
Module 6221A-SRC												

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Eehernet RGMII

Note: It can only choose between FEPHY and GEPHY

GMAC_TXD0_M1
GMAC_TXD1_M1
GMAC_TXD2_M1
GMAC_TXD3_M1
GMAC_TXCTL_M1
GMAC_TXCLK_M1

GMAC_RXD0_M1
GMAC_RXD1_M1
GMAC_RXD2_M1
GMAC_RXD3_M1
GMAC_RXCTL_M1
GMAC_RXCLK_M1

CLK_OUT_ETHERNET_M1

GMAC_MDC_M1
GMAC_MDIO_M1
GEPHY_RST_3V3IO

PHY Clock

Option2

Y6600
PHY0_XTALOUT — X1 GND 4 / 2 GND X2 3 — PHY0_XTALIN
NC/25MHz
CRY4_3R20X2R50X0R80_T
C6605 NC/18pF C0402
C6606 NC/18pF C0402
R6612 0R R0402

Option1

CLK_OUT_ETHERNET_M1 — R66140R R0402 — PHY0_XTALOUT

MAC -----> PHY

Close to PHY

PHY0_RXD0/RXDLY	R6640 22R R0402	GMAC_RXD0_M1	
PHY0_RXD1/TXDLY	R6642 22R R0402	GMAC_RXD1_M1	
PHY0_RXD2/PLLOFF	R6643 22R R0402	GMAC_RXD2_M1	
PHY0_RXD3/PHYAD0	R6644 22R R0402	GMAC_RXD3_M1	
PHY0_RXCLK/PHYAD1	R6645 22R R0402	GMAC_RXCLK_M1	C6634 NC C0402 2
PHY0_RXDV/PHYAD2	R6646 22R R0402	GMAC_RXCTL_M1	

TP6600 TP_0.7
VDD10_PHY
R66252.49K R0402 1%
VCC3V3_PHY
RSET
GMAC0_INT/PMEB
R66284.7K R0402 — VCC3V3_PHY

OD,pull up 3.3V
~~Connect to the GPIO in~~ SOC 3.3V power domain.

U6601

E_PAD	41		
AVDD33	40		
RSET	39		
AVDD10	38		
XTAL_OUT/EXT_CLK	37	PHY0_XTALOUT	
XTAL_IN	36	PHY0_XTALIN	
CLKOUT	35	PHY0_CLKOUT125_LDO1	
LED2/CFG_LDO1	34	PHY0_LED2/CFG_LDO1	
LED1/CFG_LDO0	33	PHY0_LED1/CFG_LDO0	
LED0/CFG_EXT	32	PHY0_LED0/CFG_EXT	
INTB/PMEB	31		
MDI[0]+	1	PHY0_MDI0+	
MDI[0]-	2	PHY0_MDI0-	
AVDD10	3	VDD10_PHY	
MDI[1]+	4	PHY0_MDI1+	
MDI[1]-	5	PHY0_MDI1-	
MDI[2]+	6	PHY0_MDI2+	
MDI[2]-	7	PHY0_MDI2-	
AVDD10	8	VDD10_PHY	
MDI[3]+	9	PHY0_MDI3+	
MDI[3]-	10	PHY0_MDI3-	
REG_OUT	30	PHY0_REG_OUT	
DVDD33	29	VCC3V3_PHY	
DVDD_RG	28	VCCIO_PHY	
RXC/PHYAD1	27	PHY0_RXCLK/PHYAD1	
RXCTL/PHYAD2	26	PHY0_RXDV/PHYAD2	
RXD0/RXDLY	25	PHY0_RXD0/RXDLY	
RXD1/TXDLY	24	PHY0_RXD1/TXDLY	
RXD2/PLLOFF	23	PHY0_RXD2/PLLOFF	
RXD3/PHYAD0	22	PHY0_RXD3/PHYAD0	
DVDD10	21	VDD10_PHY	
AVDD33	11		
PHYRSTB	12	PHY0_RSTN	
MDC	13	GMAC_MDC_M1	
MDIO	14	GMAC_MDIO_M1	
TXD3	15	GMAC_TXD3_M1	
TXD2	16	GMAC_TXD2_M1	
TXD1	17	GMAC_TXD1_M1	
TXD0	18	GMAC_TXD0_M1	
TXCTL	19	GMAC_TXCTL_M1	
TXC	20	GMAC_TXCLK_M1	

YT8531C-CA/YT8211F-CG
QFN40_5R00X5R00X0R90_T

VCC3V3_PHY
C6630 NC/3pF C0402 2
R66398K R0402 — VCCIO_PHY

U6600 H5007 T-24P-FT50M1L

PHY0_MDI3-	R6600 0R R0402	PHY0_MDI 3-	1	TCT1 MCT1	24 LAN0_MCT1
PHY0_MDI3+	R6601 0R R0402	PHY0_MDI 3+	2	TD1+ MX1+	23 LAN0_D-
			3	TD1- MX1-	22 LAN0_D+
PHY0_MDI2-	R6602 0R R0402	PHY0_MDI 2-	4	TCT2 MCT2	21 LAN0_MCT2
PHY0_MDI2+	R6603 0R R0402	PHY0_MDI 2+	5	TD2+ MX2+	20 LAN0_C-
			6	TD2- MX2-	19 LAN0_C+
PHY0_MDI1-	R6604 0R R0402	PHY0_MDI 1-	7	TCT3 MCT3	18 LAN0_MCT3
PHY0_MDI1+	R6605 0R R0402	PHY0_MDI 1+	8	TD3+ MX3+	17 LAN0_B-
			9	TD3- MX3-	16 LAN0_B+
PHY0_MDI0-	R6606 0R R0402	PHY0_MDI 0-	10	TCT4 MCT4	15 LAN0_MCT4
PHY0_MDI0+	R6607 0R R0402	PHY0_MDI 0+	11	TD4+ MX4+	14 LAN0_A-
			12	TD4- MX4-	13 LAN0_A+

C6600 10nF X5R25V C0402
C6601 10nF X5R25V C0402
C6602 10nF X5R25V C0402
C6603 10nF X5R25V C0402

High-tension area

R6608 75R R0805
R6609 75R R0805
R6610 75R R0805
R6611 75R R0805

J6600 RJ45_8P8C-B1DPW-3056D

LAN0_D-	8	D-	
LAN0_D+	7	D+	
LAN0_B-	6	B-	
LAN0_C-	5	C-	
LAN0_C+	4	C+	
LAN0_B+	3	B+	
LAN0_A-	2	A-	
LAN0_A+	1	A+	
		YLED+	9 LAN0_YLED+
		YLED-	10 LAN0_YLED-
		GLED+	11 LAN0_GLED+
		GLED-	12
		NC1	15
		NC2	16
		G1	13
		G2	14

Tab down

C6604 1nF 2kV C1206
LAN0_CHASSIS_GND

R6613 1M R1206
C6607 1nF 1kV C1206

RGMII Power Source	CFG_EXT	CFG_LDO[1:0]	PHY0_LED2
External 3.3V (default)	1'b1	2'b00	String 510R to GND
External 1.8V	1'b1	2'b10	String 510R to VCC3V3_PHY
Internal 1.8V	1'b0	2'b10	String 510R to VCC3V3_PHY

LAN LED

R6615 510R R0402 — LAN0_YLED+ — R6616 NC/0R R0402 — VCC3V3_PHY
PHY0_LED2/CFG_LDO1 — R6617 NC/510R R0402 — LAN0_YLED- — R6618 0R R0402
C6616 NC/100pF C0402
Reserve for EMI.

PHY0_LED1/CFG_LDO0 — R6624 510R R0402 — LAN0_GLED+
C6617 NC/100pF C0402
Reserve for EMI.

R6619 NC/4.7K R0402 — PHY0_LED0/CFG_EXT — R66204.7K R0402 — VCC3V3_PHY
R66214.7K R0402 — PHY0_LED2/CFG_LDO1 — R6622 NC/4.7K R0402
R66234.7K R0402 — PHY0_LED1/CFG_LDO0

VCC_PHY0_IO Voltage Config

R6626 NC/4.7K R0402 — PHY0_RXD2/PLLOFF — R66274.7K R0402 — VCCIO_PHY

Pull-up to disable PLL @ ALDPS mode(Low power mode)

R6629 NC/4.7K R0402 — PHY0_RXD0/RXDLY — R66304.7K R0402 — VCCIO_PHY

Pull-up for additional 2ns delay to RXC for data latching

R6631 4.7K R0402 — PHY0_RXD1/TXDLY — R6632 NC/4.7K R0402 — VCCIO_PHY

Pull-up for additional 2ns delay to TXC for data latching

R6635 NC/4.7K R0402 — PHY0_RXD3/PHYAD0 — R66344.7K R0402 — VCCIO_PHY
R6636 4.7K R0402 — PHY0_RXCLK/PHYAD1
R6637 4.7K R0402 — PHY0_RXDV/PHYAD2

PHY Address	PHYAD[2:0]
1 (default)	3'b001

PHY Address Config

VDD10_PHY
PHY0_REG_OUT — L66001.6A 2.2uH0.06ohm IND_303015
C6624 4.7uF X5R6.3V C0402
C6625 100nF X5R25V C0402
C6626 100nF X5R25V C0402
C6627 100nF X5R25V C0402
C6628 100nF X5R25V C0402
C6629 100nF X5R25V C0402

Close to PIN30 *Close to PIN21* *Close to PIN3,8,38*

PHYRSTB is 3.3V IO

GEPHY_RST_3V3IO — R6641 0R R0402 — PHY0_RSTN
C6633 100nF X5R25V C0402
Close to PIN12

VCC_3V3 — R66330R R0402 — VCCIO_PHY
C6619 4.7uF X5R6.3V C0402
C6618 1uF X5R6.3V C0402
C6620 100nF X5R25V C0402
Close to PIN28

VCC_3V3 — R66380R R0402 — VCC3V3_PHY
C6621 4.7uF X5R6.3V C0402
C6622 100nF X5R25V C0402
C6623 100nF X5R25V C0402
Close to PIN11,40

From system Power

C6631 4.7uF X5R6.3V C0402
C6632 100nF X5R25V C0402
Close to PIN29

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Audio Output

DSM_AUD_RN
DSM_AUD_RP
SPK_CTRL
SPKP_OUT
SPKN_OUT

VCC_3V3
R7014 10K

HP_OUT — R7012 0R — HPR_OUT
HP_OUT — R7015 0R — HPL_OUT
HP_DET_L_GPIO3_A1

3.3V
short @ Plug IN

JP7000
PJ6_B2323B
PJ6_B2323B
G
R
L
R
G
L IN+
DETA
DETB
DET

ED7004 ESD5451N ESD0402
ED7005 ESD5451N ESD0402
ED7006 ESD5451N ESD0402

VCC5V0_SYS
VCC5V0_PA
FB7000 120R-100M L0603 1.3A 25%
C7000 100nF X5R25V C0402
C7001 22uF X5R10V C0805

DSM OUT Filter

DSM_AUD_RN — R7002 1K R0402 — R7003 0R R0402
DSM_AUD_RP — R7005 1K R0402 — R7006 0R R0402
C7003 4.7nF X5R50V C0402
C7004 NC C0402
C7005 10nF X5R25V C0402
C7006 10nF X5R25V C0402
C7007 4.7nF X5R50V C0402
C7008 NC C0402

R7000 100K R0402 — SPK_CTRL
C7002 1uF X5R6.3V C0402
R7001 100K R0402
R7007 100K R0402

U7000
EN OUTN
VREF GND
INN VDD
INP OUTP
TCS7191A_MH
MSOP8_3R10X3R10X1R10

AUDIO BEAD
FB7001 120R-100M 1.3A 0.25 L0603
FB7002 120R-100M 1.3A 0.25 L0603
VCC5V0_PA

SPKN_OUT — J7000 (-)
SPKP_OUT — J7000 (+)
AUDIO_3PK_Jack_2P_V CN2ML_2R00_V_P_DIP

C7009 680pF C0G50V C0402
C7010 680pF C0G50V C0402
ED7000 ESD5451N ESD0402
ED7001 ESD5451N ESD0402

GAIN=576K/(Ri+6K)=7
Fc=196Hz

Note

Diffrential/Balanced Audio Signals!
Layout must be routing by differential

SPKP_OUT1
SPKN_OUT1
J5
NC/2P2.0

C23 680pF C0G50V C0402
C24 680pF C0G50V C0402
ED3 ESD5451N ESD0402
ED4 ESD5451N ESD0402

DSM_AUD_RN — R7 1K R0402
DSM_AUD_RP — R9 1K R0402
C25 4.7nF X5R50V C0402
C26 10nF X5R25V C0402
C30 10nF X5R25V C0402
C27 4.7nF X5R50V C0402
C28 10uF X5R25V C0402
HP_OUT

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Audio Input

MIC0_P
MIC0_N
MIC1_P
MIC1_N
SPKP_OUT
SPKN_OUT

ECM

AVDD1V8_AUDIO

R7100 100R R0402

C7100 4.7uF X5R6.3V C0402
R7102 1.1K R0402
MIC7100 MIC-4020 MIC2_4020

Differential Routing

C7102 NC/10pF C0402

MIC1_N

R7104 1.1K R0402
ED7100 ESD5451N ESD0402
ED7101 ESD5451N ESD0402

J8 NC/2P2.0

Note

Diffrential/Balanced Audio Signals!
Layout must be routing by differential

Analog MEMS

Option1

AVDD1V8_AUDIO

R7101 100R R0402

C7101 10uF X5R10V C0603

Differential Routing

C7103 100pF C0G50V C0402

MIC0_N

R7103 0R R0402
Close to MIC

ED7102 ESD5451N ESD0402

MIC7101
VDD
OUT
GND1
GND2
MEMS Analog MIC Top Type
MIC_3SM121MZT1UA_MIC4_2R95X3R76X1R10

Note

Diffrential/Balanced Audio Signals!
Layout must be routing by differential

Loopback

Option2

MIC0_P

R71053 3.3K R0402 1%

C7104 2.2nF X5R50V C0402
C7107 2.2nF X5R50V C0402

C7105 22nF X5R16V C0402
C7108 22nF X5R16V C0402

R7108 390R R0402 1%
R7111 390R R0402 1%

R71062 2.2K R0402 1%

SPKP_OUT1

MIC0_N

R71123 3.3K R0402 1%

R71132 2.2K R0402 1%

SPKN_OUT1

Differential Routing

Differential Routing

Note

Diffrential/Balanced Audio Signals!
Layout must be routing by differential

Line In

Option3

Vrms<1.0V

MIC0_P

R71071 8K R0402

J7 NC/2P2.0

C7106 NC/10pF C0402

MIC0_N

R7114 0R R0402
Close to SoC

R7110 390R R0402

R7109 0R R0402
Close to Connecter

ED7103 ESD5451N ESD0402

Note

Diffrential/Balanced Audio Signals!
Layout must be routing by differential

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Key Array

SARADC0_IN0_KEY

RESET

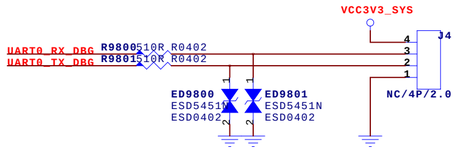
Option: If don't use, it can be delete.



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Debug UART

UART0_RX_DBG
UART0_TX_DBG



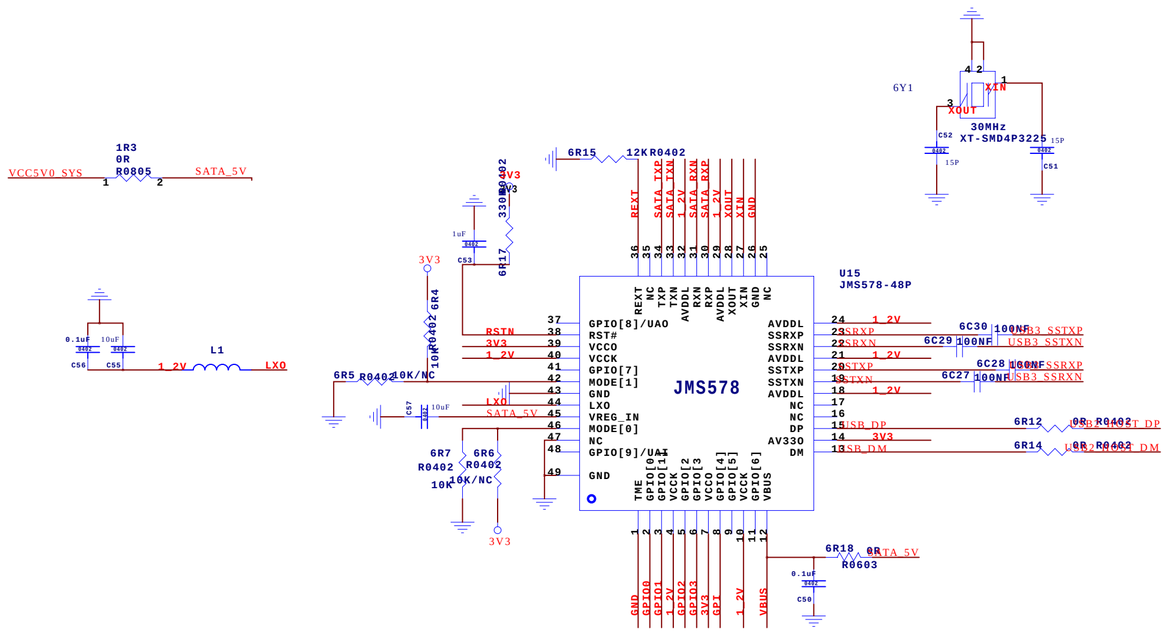
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HOLE

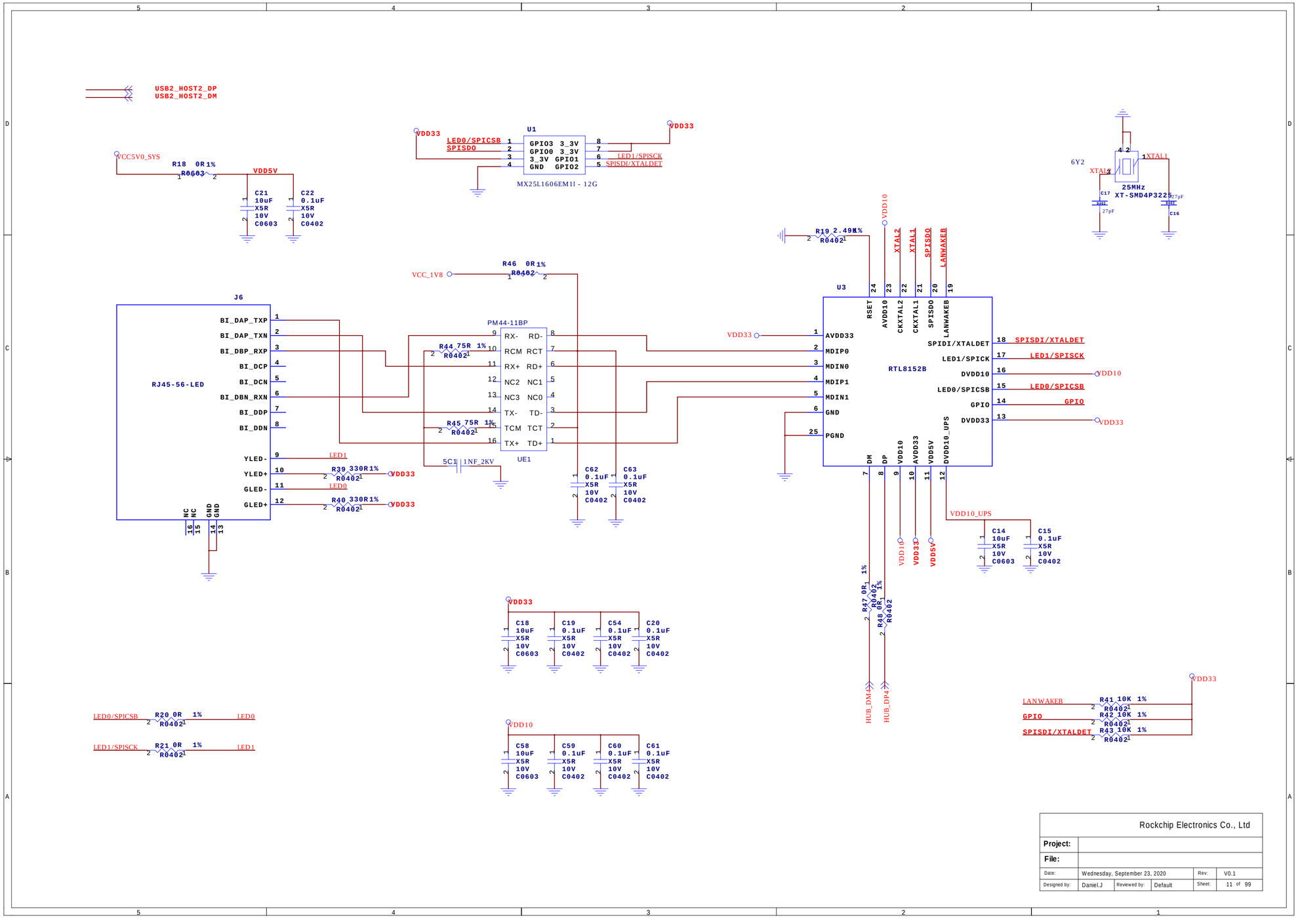
MARK



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AMedia			
Title: USB3.0 HUB			
Size: C	Document Number:	Drawn:	Rev: V01
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USB2_HOST2_DP
USB2_HOST2_DM
VCC5V0_SYS
R18 0R 1%
R0603
VDD5V
C21
10uF
X5R
10V
C0603
C22
0.1uF
X5R
10V
C0402
VDD33
LED0/SPICSB
SPISDO
U1
GPIO3 3_3V
GPIO0 3_3V
3_3V GPIO1
GND GPIO2
LED1/SPISCK
SPISDI/XTALDET
MX25L1606EM1I - 12G
6Y2
XTAL2
XTAL1
25MHz
XT-SMD4P3225
27pF
C17
C16
VCC_1V8
R46 0R 1%
R0402
J6
BI_DAP_TXP
BI_DAP_TXN
BI_DBP_RXP
BI_DCP
BI_DCN
BI_DBN_RXN
BI_DDP
BI_DDN
RJ45-56-LED
YLED-
YLED+
GLED-
GLED+
NC
NC
GND
GND
LED1
LED0
R39 330R1%
R40 330R1%
PM44-11BP
RX- RD-
RCM RCT
RX+ RD+
NC2 NC1
NC3 NC0
TX- TD-
TCM TCT
TX+ TD+
UE1
R44 75R 1%
R45 75R 1%
5C1 1NF_2KV
C62
0.1uF
X5R
10V
C0402
C63
0.1uF
X5R
10V
C0402
VDD10
R19 2.49K 1%
R0402
U3
RSET
AVDD10
CKXTAL2
CKXTAL1
SPISDO
LANWAKEB
AVDD33
AVDD0
MDIP0
MDIN0
MDIP1
MDIN1
GND
PGND
RTL8152B
SPIDI/XTALDET
LED1/SPICK
DVDD10
LED0/SPICSB
GPIO
DVDD33
DVDD10_UPS
DM
DP
VDD10
AVDD33
VDD5V
VDD10_UPS
C14
10uF
X5R
10V
C0603
C15
0.1uF
X5R
10V
C0402
R47 0R 1%
R48 0R 1%
R0402
HUB_DM4
HUB_DP4
C18
10uF
X5R
10V
C0603
C19
0.1uF
X5R
10V
C0402
C54
0.1uF
X5R
10V
C0402
C20
0.1uF
X5R
10V
C0402
C58
10uF
X5R
10V
C0603
C59
0.1uF
X5R
10V
C0402
C60
0.1uF
X5R
10V
C0402
C61
0.1uF
X5R
10V
C0402
LED0/SPICSB
R20 0R 1%
LED0
LED1/SPISCK
R21 0R 1%
LED1
LANWAKEB
GPIO
SPISDI/XTALDET
R41 10K 1%
R42 10K 1%
R43 10K 1%
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Project:
File:
Date: Wednesday, September 23, 2020
Rev: V0.1
Designed by: Daniel.J
Reviewed by: Default
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